[Diabetes Dataset]

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1. Problem Statement and Background

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Statement of the Problem:

This project aims to develop a binary classification model to predict whether a female patient of Prima Indian Heritage has diabetes. The goal is to use a set of health-related variables to accurately classify patients as either diabetic or non-diabetic. This problem is important because early detection of diabetes can lead to better management and prevention of the disease, improving patient outcomes and reducing the long-term health risks associated with diabetes.

Dataset Overview:

The dataset used in this project is hosted on Kaggle and originates from the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK). It is a subset of a larger dataset, specifically focused on female patients of Prima Indian Heritage, aged 21 and older. The dataset contains several independent variables, such as medical and health metrics, that are used to predict the dependent variable: whether the patient has diabetes or not. The target is binary: 1 for diabetes, 0 for non-diabetic.

Success Measures:

The primary success measure for this project will be the accuracy of the predictive model in classifying the diabetes status of patients. This will be evaluated by comparing the model's predictions to the known outcomes in the dataset. While other metrics (e.g., precision, recall, ROC) may be useful, the focus here will be on overall accuracy as a proxy for the model's ability to correctly classify diabetes status.

Background and Importance:

Diabetes is a chronic and increasingly prevalent condition that affects millions of people worldwide. It is particularly concerning in the U.S., where diabetes-related complications can significantly reduce quality of life and lead to higher healthcare costs. Current healthcare systems often focus on reactive treatment after diagnosis, but early detection of diabetes could help shift the focus to preventative care. Accurate predictive models can assist healthcare professionals in identifying at-risk individuals early, enabling interventions that can delay or prevent the onset of diabetes and its associated complications.

Related Work:

There has been significant work in the area of diabetes prediction using machine learning, particularly in classifying diabetic versus non-diabetic patients based on medical and lifestyle factors. Various models, including logistic regression, decision trees, and neural networks, have been applied to similar datasets to predict diabetes risk. However, the focus on a specific demographic (Prima Indian Heritage women) provides an opportunity to refine models that might be more tailored to this group's unique health characteristics.

Conclusion:

By developing a model that accurately predicts diabetes status, this project seeks to contribute to the broader effort of improving early detection methods for diabetes, which could have significant implications for patient care and public health strategies.

2. Data and Exploratory Analysis

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The Diabetes Dataset is hosted on Kaggle and originated from the National Institute of Diabetes and Digestive Kidney Diseases. The dataset is a subset of the larger original dataset by using constraining factors to reduce the selection of variable instances. All of the patients within the dataset are females of Prima Indian heritage and are at least 21 years old. There are several independent medical predictor variables and one target outcome (dependent) variable, whether the patient has diabetes or not.

Dataset Variables:

1. Pregnancies: To express the Number of pregnancies
2. Glucose: To express the Glucose level in blood
3. BloodPressure: To express the Blood pressure measurement
4. SkinThickness: To express the thickness of the skin
5. Insulin: To express the Insulin level in blood

6. BMI: To express the Body mass index
7. DiabetesPedigreeFunction: To express the Diabetes percentage
8. Age: To express the age
9. Outcome: To express the final result 1 is Yes and 0 is No

General data cleaning included outlier removal and missing value imputation, in this dataset the missing values were 0 (excluding pregnancies and diabetic outcome), which is unrealistic for a living patient in any of the above variables.

Libraries Utilized: Tidyverse, readxl, caret, pROC, mice, caret, and corrplot.

Data Cleaning and Preparation

The dataset used in this project required several steps of cleaning to ensure its suitability for predictive modeling. The main issues addressed during the cleaning process were missing values, outliers, and the treatment of unrealistic data entries. Below is a detailed discussion of the steps taken to clean and preprocess the data, as well as the tools and methods used in the process.

Handling Missing Values:

One of the primary challenges in this dataset was the presence of missing values, particularly in columns such as *Insulin* and *BloodPressure*. In this dataset, missing values were represented by zero (0) for most variables (with the exception of the *Pregnancies* column). Since a value of zero is unrealistic for these variables in the context of a living patient (e.g., having zero insulin levels or zero blood pressure is not a plausible scenario), these entries were treated as missing data rather than valid observations.

Two methods of imputation were used to address these missing values:

1. **Mean Imputation with Clustering for Insulin:** For the *Insulin* column, a clustering-based imputation strategy was employed. Since insulin levels are likely influenced by other medical metrics, particularly *Glucose*, we used clustering techniques to group patients into classifications based on glucose levels (ranging from low to high). The mean insulin values within each glucose group were then used to replace missing insulin values, under the assumption that insulin levels would correlate with glucose levels.
2. **Multivariate Imputation by Chained Equations (MICE):** The MICE method was applied to impute missing values across other columns in the dataset, except for *Pregnancies* and *Insulin*, which were handled differently. MICE is a more sophisticated technique that generates multiple imputed values for each missing entry by utilizing the relationships between variables in the dataset. This method was preferred for columns where the relationship between variables was complex and could benefit from imputations based on multiple features in the dataset.

After imputation, both cleaned datasets (one with clustering-based imputation and one with MICE) were used to train logistic regression models to predict the likelihood of diabetes, with the goal of evaluating which cleaning method produced a model with higher predictive accuracy.

Outlier Detection and Handling:

In addition to missing values, several variables in the dataset contained outliers that needed to be addressed:

- The *BloodPressure* column had 45 outlier values, which seemed abnormally high or low compared to the rest of the dataset.
- The *DiabeticPedigreeFunction* had 27 outliers.
- The *Insulin* column, which already had missing values, also contained 26 outliers.
- The *BMI* column had 18 outliers.

These outliers were identified using simple statistical methods (e.g., identifying values beyond 1.5 times the interquartile range). In cases where the outliers appeared to be due to data entry errors or extreme values that were not physiologically plausible, these were either capped or removed, in terms of that.

Related Data Variables:

Looking at the correlation matrix of data we can see various relationships between features within the dataset (See visualization matrix in MICE R script). Glucose and Outcome show a moderately strong positive correlation, suggesting higher glucose levels results in a positive diabetic diagnosis, which we would expect given blood glucose levels is a direct biological link to glucose cases. BMI and SkinThickness show a strong positive correlation, this is also expected as skinfold thickness can be used to estimate body fat, which links to BMI. Insulin and Glucose has a strong positive correlation, this is also expected as high glucose levels are generally accompanied by elevated insulin levels. Pregnancies also show a moderately positive correlation with age, naturally older people are more likely to have more children.

Data Preprocessing Tools and R Code:

To perform the cleaning, we used the following tools and R packages:

- **Tidyverse** for data manipulation and imputation tasks.
- **mice** package for performing Multivariate Imputation by Chained Equations (MICE) for missing values across the dataset.
- **ggplot2** and **corrplot** for visualizing data distribution, correlations, and variable relationships.
- **Caret** and **pROC** was used for initial model training, cross-validation, and performance measure calculations using confusion matrices and AUC and ROC curves.
- **Readxl** used to import data from the CSV diabetes dataset.

Conclusion:

The cleaning process involved addressing missing values, outliers, and unrealistic data entries to ensure the dataset was ready for analysis. The imputation methods used (clustering and MICE) provided a way to fill in missing data in a reasonable manner, and outlier detection helped mitigate any potential issues that could arise from extreme values. The cleaned dataset was then used to build predictive models and assess their accuracy. Further evaluations will focus on comparing the performance of models built with different imputation methods to determine the best approach for this specific dataset.